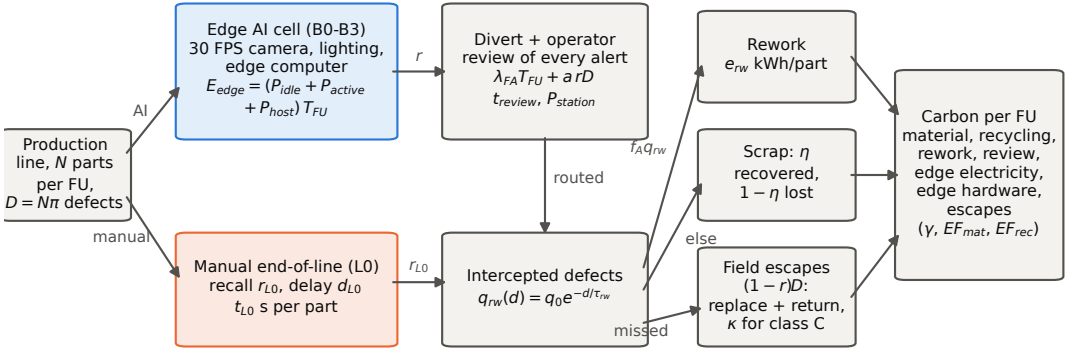
Production line, N parts per FU, $D = N\pi$ defects
AI
manual
Edge AI cell (B0-B3)
30 FPS camera, lighting, edge computer
$E_{edge} = (P_{idle} + P_{active} + P_{host})\, T_{FU}$
Manual end-of-line (L0)
recall r_{L0}, delay d_{L0}
t_{L0} s per part
r
r_{L0}
Divert + operator review of every alert
$\lambda_{FA} T_{FU} + a\, r D$
t_{review}, $P_{station}$
routed
Intercepted defects
$q_{rw}(d) = q_0 e^{-d/\tau_{rw}}$
$f_A q_{rw}$
else
missed
Rework
e_{rw} kWh/part
Scrap: η recovered, $1-\eta$ lost
Field escapes
$(1-r)D$: replace + return, κ for class C
Carbon per FU
material, recycling, rework, review, edge electricity, edge hardware, escapes
(γ, EF_{mat}, EF_{rec})